Project Title: MediSync: Smart Patient Triage & Risk Analyzer

Domain: Healthcare AI / Computational Biology (CB)

Abstract:

In high-pressure healthcare environments, the manual triage of patients based on vital signs leads to critical delays. "MediSync" is an Object-Oriented Java-based application designed to automate this process using Agentic AI principles. The system ingests a real-world dataset of 2,000 patients from Kaggle (Healthcare Disease Prediction Dataset) using Java File I/O (BufferedReader). It utilizes a modular rule-based AI engine (Strategy Pattern) to analyze patient vitals—including Systolic Blood Pressure, Glucose Levels, and BMI—and dynamically routes them to the appropriate department (Cardiology or Endocrinology).

The architecture heavily implements core OOP principles such as:

Inheritance: Patient extends the abstract Person class.

Polymorphism & Interfaces: A MedicalAnalyzer interface is implemented by separate CardioAnalyzer and DiabetesAnalyzer classes.

Design Patterns: The AIEngine uses a Factory Pattern to dynamically assign the correct analyzer.

Collections & Sorting: A custom Comparator sorts the ArrayList of patients, prioritizing high-risk individuals.

Exception Handling: A custom InvalidVitalException prevents corrupted data from entering the system.

MediSync provides a scalable, robust solution for modern hospital management, demonstrating how OOP can solve complex real-world medical data challenges.